

# Chapter 6: Conclusion and Future Scope

## 6.1 Conclusion

Medical Imaging Assistant: AI-Powered advances the application of Deep Learning Technology for the everyday practice of Neuro-Oncology. The project has moved from an initiative to diminish delays/errors associated with brain tumors' diagnostic evaluation into an entire quality product having the capability to assist in clinical practice.

Utilizing a robust implementation of the U-Net architecture and the finely-tuned EfficientNet B4 model classifier we have been able to create a seamless/automated pipeline that simultaneously cleans noise through enhancement, sharpens structural information detail, and creates a multi-tumor classification prediction. The final trained model produced predictions with over 97.9% in test accuracy tested against a large data set that was not introduced into the training data set or 97.9% fewer cases misdiagnosed or missed entirely, while running well on standard computing hardware.

This project provides radiologists with an easily accessible, pre-packaged Gradio Interface, which can be uploaded and run via a one-line command. The software outputs can be viewed using any current web-browser, allowing succinct integration into ongoing healthcare imaging processes, with minimal collaboration required from their IT departments. To summarize, state-of-the-art methods for analyzing and interpreting medical images are no longer just for the well-endowed research institutions and cloud based supercomputers. With a little thoughtful architecture, consistent pre-processing, and the creation of usable deployment strategies, it is possible to provide effective, clinically significant medical diagnostic tools via Artificial Intelligence (AI) to every radiological imaging hub and beyond, thereby coming one step closer to faster and more accurate diagnoses of brain tumors for anyone, anywhere.

### 6.1.1 Summary of Key Findings

Our discoveries from this research go beyond simply identifying which model is better, but also provide insight into what factors contribute to a medical imaging system's reliability and practicality.

Preprocessing Is Essential. The series of enhancement steps (DnCNN  $\rightarrow$  EDSR  $\rightarrow$  U-Net) is not an optional extra; it is critical to the difference between a poor performing classifier and an extremely high performing classifier. Each step in the preprocessing pipeline (removing scanner noise, recovering small details, and isolating clean brain tissue) enables the classifier to only look at the clinically relevant information in the image. Removing or failing to do any of the enhancement steps caused a substantial decrease in classification accuracy and increased the number of false positives.

High "No Tumor" Recall Ensures Clinical Safety. One of the most significant results is that the system has a recall rate of  $\approx 99\%$  for the No Tumor subclass. Practically speaking, this means that a model almost never incorrectly identifies a healthy patient to have a tumor a false positive that can result in much of the anxiety and stress associated with unnecessary biopsies. To earn the trust of clinicians, any model must be able to provide a level of reliability on this level.

Generalized Strongly with Variability in Datasets. As a result of normalizing aggressively through the Pre-processing Chain, the differences between Scanner Types, Acquisition Parameters (e.g., MR Scans) and Image Quality to a much greater degree than originally anticipated, the resulting Classifier also generalizes far better than we expected. In fact, even when Testing the test set using scans from Centers/Protocols that were not represented in the training dataset, the performance of the Classifier remained constant. This robustness is what primarily distinguishes Lab Prototype systems from Production-grade systems that can be implemented/installed in hospitals with varying infrastructures.

Additionally, this highlights an important lesson for medical imaging: while raw model capacity is important, what is more important in medical imaging is the consideration of Domain-Specific Preprocessing, as well as a rigorous analysis of Clinical Failure Modes. These elements combined enabled the conversion of a very successful research finding into an implementation where Radiologists could begin using it tomorrow.

### **6.1.2 Contributions to the Field**

This paper presents advancements in medical imaging technology through three major contributions:

First, we demonstrate that the accuracy achieved by developing a slice-based processing algorithm and then enhancing its performance through targeted application of super-resolution on a single slice at a time, applying a large number of different augmentations to increase the amount of training data available, and then using multiple methods for enhancing performance (i.e. cascaded enhancement) comes close to the level of accuracy achieved by using a far more complex and resource-demanding 3D volumetric approach. An increased number of penetrating and widespread hospitals and clinics will benefit from applying this algorithm without having to spend money on high powered computers for image analysis.

Second, rather than simply providing the trained models created by our work, we have built an entire web-based system that includes a fully-functioning dashboard that automates the procedures for adding an image and patient meta-data; real-time processing and analysis of an image; and providing side by side views (two screens) of an interpreted image and a process to produce an image-what the physician would need to show to his/her patient to discuss their results.

By creating this system in this way, we will help to overcome the "last mile" challenge in medical artificial intelligence, where medical professionals have moved from research level use of technology to daily clinical practice.

The final model has been trained and validated against over 7000 MRI scans obtained from various public sources (e.g., Kaggle, Figshare), which introduces variability into our model due to differing protocol usage, scanner brands, and patient demographics; thus providing much greater generalization capability than the majority of previous research work based on well controlled, single institution data sets. Therefore, we are extremely confident that our model will generalize to the diverse nature of the clinical environment where it will eventually be implemented.

The work outlined here represents substantial contributions to the field both from a theoretical standpoint and through the development of a functional tool that can realistically improve brain tumor diagnosis for patients in the near future.

### **6.1.3 Limitations of the Current System**

Even though the model has performed well, there remain several evident limitations of the current model that indicate areas for improvement. They are:

- **Absence of 3D spatial context** The model is unable to capture 3D volumetric information because it treats each MRI slice separately. As a result, significant and relevant volumetric features including tumor size, shape (e.g., exact size and volume from the slices), growth pattern(s) (i.e., the operation of the tumor across consecutive slices), and possible pattern(s) of infiltrative tumor growth into surrounding normal tissue, etc. will be absent from predictions. Implementation of true 3D convolutions or slice-based aggregation techniques would create a means to capture useful 3D information and also create additional possible clinical outputs (for example, use of predictions to create automated tumor volume estimates).
- **Limited explainability** Although EfficientNetB4 is capable of producing accurate patient-specific illness diagnostic placement, it operates in a 'black box' fashion, meaning that, generally speaking, Clinician's want to understand why a specific prediction was provided. Therefore, the addition of new interpretability capabilities, such as Grad-CAMs, attention-based techniques, and/or concept-based interpretability tools would greatly improve confidence in results and make it much easier for regulatory and compliance authorities to evaluate and assess the model.
- **Performance on low-field scanners remains to be tested** Currently, all evaluations of the model were conducted using MRI images from scanners operating in the 1.5 T–3 T range. Many hospitals, particularly in low-resources countries, are using scanners operating in the  $<0.5$  T range, which produce images with less signal-to-noise ratios (SNR) and lower contrast than higher-frequency scanners. By including lower T data in the training and evaluation process (or using domain adaptation methods), this model has the potential to be implemented in a significantly broader range of clinical settings.

Stating these limitations at the outset allows stakeholders to set reasonable expectations and gives stakeholders a clear path for making improvements to advance the clinical use of the system.

## **6.2 Future Scope**

By leveraging the current prototype, the planned enhancements will overcome the limitations already identified in the current prototype and will get the system much closer to its clinical usage potential.

**6.2.1 Advancement of 2D Segmentation Capabilities** Despite having effective skull stripping capabilities with the current model architecture of U-Net, our next goal is to build upon this capability by developing an improved version capable of producing accurate segmentation of tumors in each MRI slice. We will pursue advanced U-Net architectures like Attention U-Net and U-Net++ that have demonstrated superior performance in detecting contour complexity and small boundary changes between tissues. The tumor mask that will be generated for each MRI slice will be presented in a manner that allows the radiologist to see exactly how the tumor appeared and where it appeared in relation to surrounding structures at the time the MRI was obtained. By combining the individual mask images from each slice of an MRI, we will also be able to estimate the overall volume of the tumor prior to creating a true three-dimensional (3D) model through the application of all slice images.

**6.2.2 Integration of Explainable AI (XAI)** In order to mitigate the 'black box' problem associated with EfficientNetB4, we will integrate Grad-CAM++ (or another similar state-of-the-art attribution technique) into the model's inference pipeline to provide clinicians with real-time heatmaps that visualise the relative contribution to the classification decision of different regions within the scanned image. By providing clinicians with visual guidance showing where the model is concentrating its focus (or where it may have misinterpreted what it saw as a tumour due to some other cause), trust can be developed between the clinicians and the model. As a result, any error made by the model will be caught earlier and thus will reduce the number of rare errors occurring during routine clinical practice, and these visual explanations will provide an additional layer of support during both clinical audits and regulatory submission processes.

**6.2.3 Transition to 3D Volumetric Segmentation** 3D U-Net, V-Net and nnU-Net architectures are used to process 3D volumetric data. Rather than analyzing two-dimensional slices separately, these architectures analyze entire volumes of material in three-dimensional space all at once. This means that a model will be able to develop accurate spatial relationships between the individual slices of an entire volume through all three dimensions. As a result, this model will enable highly accurate 3-dimensional tumor segmentation, automated volumetric measurement, and improved classification of the internal structure

(shape) and invasion type of tumors. The ability to accurately segment tumors is necessary to provide appropriate treatment planning for surgical and radiotherapy treatment, and it is important for continuous monitoring of tumor response to therapy.

**6.2.4 Generative AI for Automated Radiology Reporting** The integration of a medical visual language model (e.g. LLaVA-Med or RadFM or GPT-4 V) will add an additional benefit. Using both the original image and structured output from our model, a radiology report will be generated by the VLM to produce complete, human-readable reports consisting of all necessary components including findings, measurements, location descriptors, and impressions. By utilizing these models, reporting styles will become standardized across all institutions, documentation workload for radiologists will decrease considerably, and radiologists will be able to devote more time to speaking with patients and working on difficult, complex cases.

**6.2.5 Frontend–Backend Integration and Clinical-Grade Deployment** We will convert the current Gradio prototype into a completely working production system that will include:

- Secure (FastAPI/Flask) API with the ability to authenticate users by roles(as needed) securely store patient information.
- The ability to accept DICOM files via upload, provide PACS integration hooks, offer the ability to deploy either on-premise or cloud (Docker/Kubernetes).
- Create a smooth transition from the current, relatively simple research prototype to a full-fledged clinical assistant.
- These are just a few of the steps we are taking to ensure the final product is regulatory compliant, fully functional, user-friendly, and actually useful for radiologists who will utilize it on a daily basis.